

CIFAR-10 Client Inference Attack: Transfer Report

Planned-run results

Scope and protocol

FedAvg is evaluated for 20 communication rounds and 5 local epochs on the non-i.i.d. CIFAR-10 split. The IN condition has three active clients; the OUT condition removes the target and has two. Results use seeds 42–44, a 10% target shadow split, noise multiplier 0.01, clipping norm 5, and both clean and Gaussian-perturbed shadow data (standard deviation 20% of the maximum pixel value). Attack scores follow the paper: $s_r = -\mathcal{L}(W^{(r)}, D_x)$. The noisy column is the paper-faithful configuration for the multi-round attack (10% split of the target’s training data perturbed at 20% of the maximum pixel value); the clean column is an additional strong-adversary reference of ours. Both shadow variants were verified to contain the same 150 target examples in the IN and OUT conditions, with an identical, deterministic noise draw, so the two conditions differ only in whether the target participates in training.

Client data distribution

This CIFAR-10 transfer experiment reuses the corrected three-client count profile from the paper’s Table 9, but maps its four columns to the CIFAR-10 classes shown below.

Table 1: Canonical examples assigned to each client. Parentheses show within-client class percentages.

Client	Role	IN	OUT	Total	airplane	automobile	bird	cat
1	Attacker	yes	yes	1747	120 (6.9%)	9 (0.5%)	1122 (64.2%)	496 (28.4%)
2	Bystander	yes	yes	1491	80 (5.4%)	11 (0.7%)	894 (60.0%)	506 (33.9%)
3	Target	yes	no	1882	524 (27.8%)	29 (1.5%)	550 (29.2%)	779 (41.4%)
All	Canonical pool			5120	724 (14.1%)	49 (1.0%)	2566 (50.1%)	1781 (34.8%)

How non-IID is it? Total-variation distance from the canonical global class distribution is 0.141 for Client 1, 0.098 for Client 2, and 0.209 for the target. Pairwise distance is only 0.058 between Clients 1 and 2, but rises to 0.350 between Clients 1 and 3 and 0.307 between Clients 2 and 3. Thus, the split is deliberately target-skewed rather than uniformly heterogeneous: the attacker and bystander are close to each other, while the target is meaningfully different. The target has 27.8% airplane and 29.2% bird examples, versus 5.4–6.9% airplane and 60.0–64.2% bird for the other clients. The automobile class remains rare everywhere (0.5–1.5%), so it contributes little to the heterogeneity signal.

Table 2: Examples actually used for local training after the deterministic stratified 80/20 split.

Client	Role	Train total	airplane	automobile	bird	cat	Local evaluation total
1	Attacker	1397	96	7	897	397	350
2	Bystander	1192	64	9	715	404	299
3	Target	1505	419	23	440	623	377

Single-round CIA

Table 3: Published Alzheimer reference (Tables 10 and 12); shown for protocol context only, not as a CIFAR-10 baseline. The target loss is measured on the clean shadow data.

Approach	Aggregated loss	Target loss	Difference (%)	Test loss
Vanilla FL	1.032	1.182	12.719	1.010
Global-DP	3.603	4.848	25.679	3.514
Metric-privacy	1.135	1.506	24.631	1.106

Table 4: Ours, first-round IN condition (mean $\pm$ sample std over three seeds).

Approach	Aggregated loss	Clean target	Clean diff. (%)	Noisy target	Noisy diff. (%)
Vanilla FL	1.435 ± 0.045	1.194 ± 0.048	-20.266 ± 6.616	1.200 ± 0.033	-19.624 ± 5.762
Global-DP	1.426 ± 0.023	1.211 ± 0.018	-17.823 ± 0.762	1.209 ± 0.003	-17.966 ± 1.942
Metric-privacy	1.426 ± 0.023	1.208 ± 0.021	-18.054 ± 0.879	1.207 ± 0.004	-18.118 ± 1.857

CIA score trajectories across communication rounds

— IN trajectory — OUT trajectory

Lines = mean across seeds 42, 43, 44; translucent bands = ± 1 sample standard deviation.

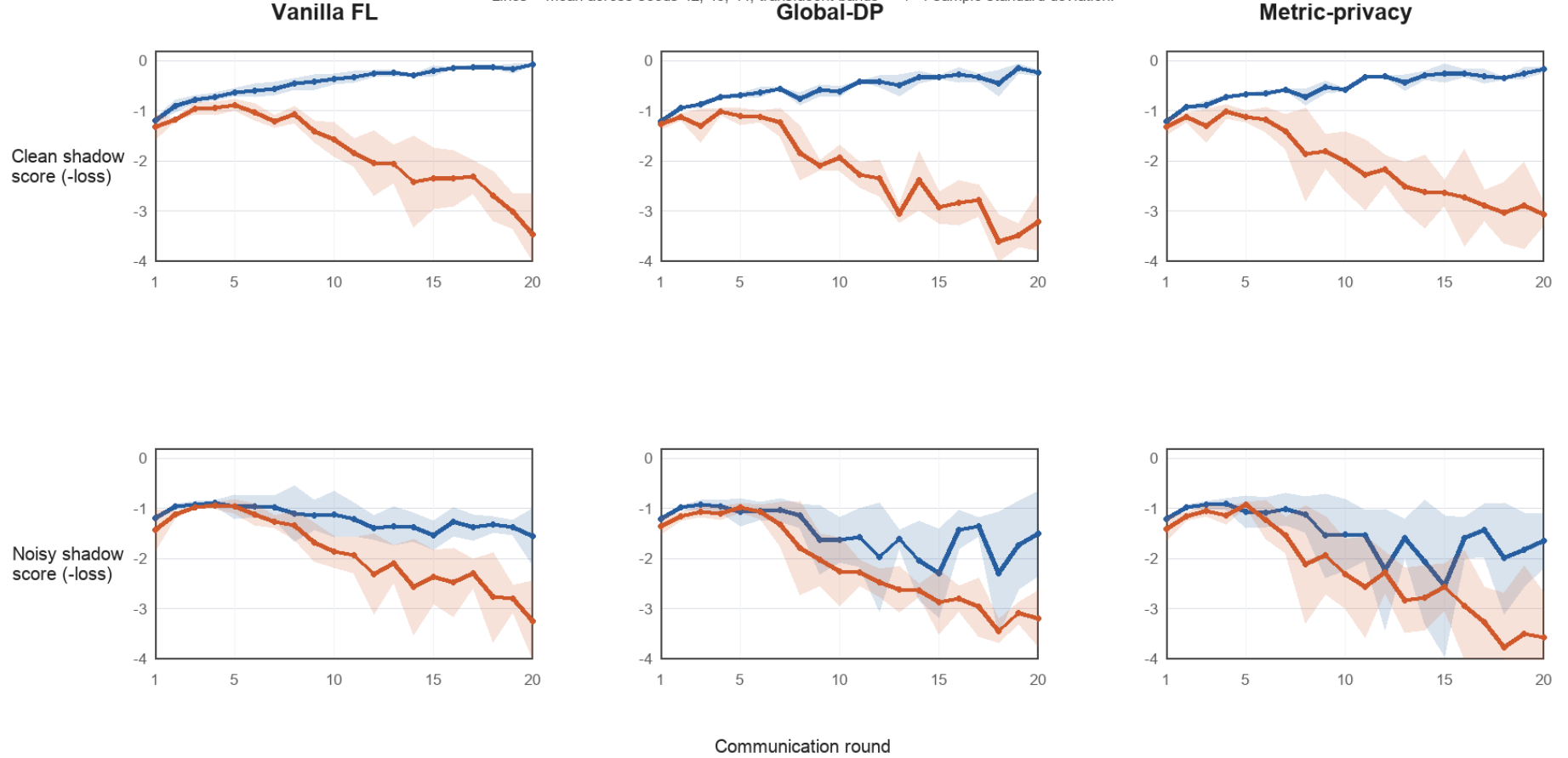


Figure 1: CIA score trajectories. Each panel uses raw scores $s_r = -\mathcal{L}(W^{(r)}, D_x)$ and the shared y-axis range $[-4, 0.2]$.

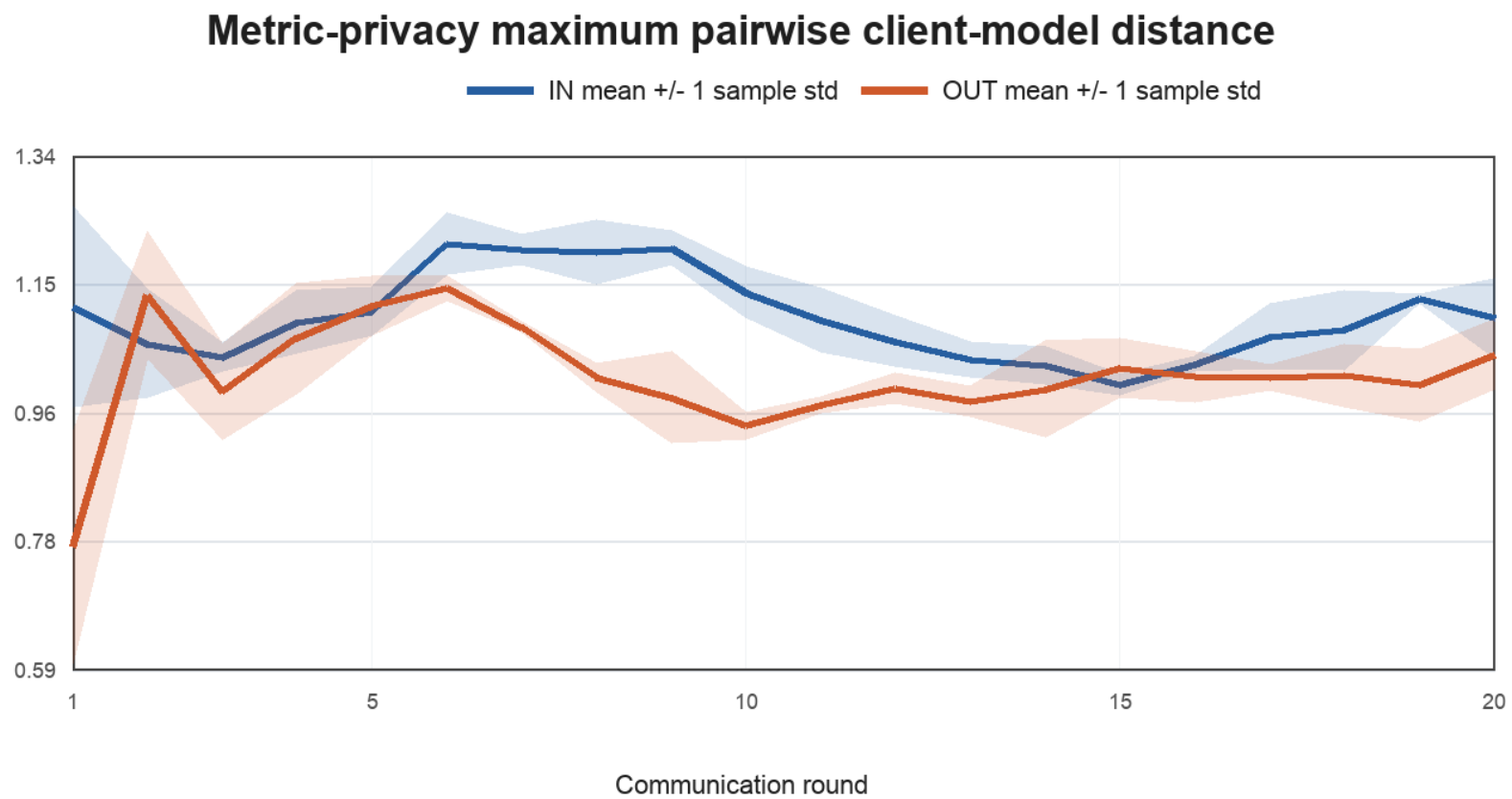


Figure 2: Metric-privacy maximum pairwise client-model distance recorded before each server aggregation.

Multi-round CIA

The directional ROC AUC estimates the probability that an IN score ranks above an OUT score. The paper reports one set of 20 IN and 20 OUT round scores. Our per-seed estimates use the same 20-versus-20 construction; the pooled summary combines all three seeds (60 IN and 60 OUT scores). Confidence intervals are deterministic 95% percentile intervals from 10,000 stratified bootstrap resamples. Those intervals resample rounds as if they were independent draws, which they are not: the 20 scores of a run form one autocorrelated trajectory and the real replication unit is the seed ($n = 3$). The reported intervals are therefore optimistic and should be read as paper-comparable, not as calibrated coverage.

Table 5: Published Alzheimer reference (Table 13); shown for protocol context only, not as a CIFAR-10 baseline. The attack uses noisy shadow data.

Approach	Attack AUC (95% CI)	Acc. IN	Acc. OUT	F1 IN	F1 OUT
Vanilla FL	0.890 (0.767, 0.977)	0.948 ± 0.007	0.894 ± 0.007	0.930 ± 0.006	0.729 ± 0.024
Global-DP	0.397 (0.224, 0.585)	0.637 ± 0.009	0.680 ± 0.026	0.426 ± 0.018	0.400 ± 0.008
Metric-privacy	0.493 (0.296, 0.689)	0.794 ± 0.023	0.746 ± 0.029	0.695 ± 0.035	0.480 ± 0.022

Table 6: Ours, pooled across seeds. Utility is mean $\pm$ sample std over rounds 16–20 and all three seeds.

Approach	Clean AUC (95% CI)	Noisy AUC (95% CI)	Acc. IN	Acc. OUT	F1 IN	F1 OUT
Vanilla FL	0.978 (0.954, 0.995)	0.769 (0.681, 0.851)	0.627 ± 0.021	0.529 ± 0.021	0.610 ± 0.026	0.477 ± 0.027
Global-DP	0.987 (0.971, 0.999)	0.714 (0.618, 0.801)	0.597 ± 0.031	0.494 ± 0.024	0.578 ± 0.042	0.445 ± 0.035
Metric-privacy	0.988 (0.971, 0.998)	0.722 (0.629, 0.808)	0.604 ± 0.024	0.507 ± 0.047	0.584 ± 0.032	0.464 ± 0.062

Interpretation. Clean-shadow access produces strong separation for every approach. For noisy-shadow access, metric-privacy and global-DP also remain far above 0.5 in this Alzheimer setup; neither reproduces the near-random multi-round behavior reported by the paper. Because this is a directional AUC, values below 0.5 indicate reversed score ordering and can still be exploitable if an attacker is allowed to invert the decision rule.

Per-seed multi-round attack results

Table 8: Paper-comparable 20-round AUC estimates for each matched IN/OUT seed pair.

Approach	Seed	Clean AUC (95% CI)	Noisy AUC (95% CI)
Vanilla FL	42	0.983 (0.940, 1.000)	0.782 (0.620, 0.910)
Vanilla FL	43	0.980 (0.930, 1.000)	0.677 (0.497, 0.843)
Vanilla FL	44	0.983 (0.943, 1.000)	0.843 (0.700, 0.948)
Global-DP	42	0.988 (0.950, 1.000)	0.853 (0.725, 0.953)
Global-DP	43	0.983 (0.943, 1.000)	0.540 (0.355, 0.725)
Global-DP	44	0.990 (0.963, 1.000)	0.725 (0.552, 0.875)
Metric-privacy	42	0.985 (0.950, 1.000)	0.868 (0.740, 0.968)
Metric-privacy	43	0.988 (0.955, 1.000)	0.568 (0.383, 0.755)
Metric-privacy	44	0.993 (0.970, 1.000)	0.775 (0.613, 0.907)

Data provenance

All “ours” values are computed directly from the CIFAR-10 IN/OUT artifacts under `results/planned_runs/cifar`. Published reference values are transcribed from Sainz-Pardo Diaz et al., *Metric-privacy-inspired noise calibration in federated learning: Improving convergence and preventing client inference attacks*, Tables 10, 12, and 13.